Sets & Functions

A "collection of objects viewed as a single entity" is called a *set*

- If the objects in the set can be counted then it's common for the set to be written by enclosing the objects in curly braces $\{...\}$
  - The curly braces are extremely important: $\{1, 4, 5\} \neq (1, 4, 5)$
- Examples of sets: $\{$Maggie, Bobby$\}$, or $\{2, 5, 3, 8, 10\}$, or $\{$red, green, blue$\}$
- The elements of a set must all be unique: $\{1, 1, 3, 4, 4\}$ is not a set (it's a *multiset*)
- The ordering of elements in a set does not matter: $\{$red, blue, green$\}$=$\{$green, blue, red$\}$

Basic and essential notation involving sets

- "Is an element of " is $\in$
  - green $\in$ {red, green, blue}
  - magenta $\notin$ {red, green, blue}
- "Is a subset of" is $\subset$
  - {green, red} $\subset$ {red, green, blue}
  - Note that green $\not\subset$ {red, green, blue} because "green" is not a set!
  - However {green} $\subset$ {red, green, blue} because {green} IS a set
  - Sometimes a distinction is made between $\subset$ (strict subset) and $\subseteq$ (weak subset); your book doesn't make the distinction

- "Union" is $\cup$, and represents the set of elements in *either* set

  - $\{$Maggie, Bobby$\} \cup \{2, 5, 3, 8, 10\}$
    $=\{$Maggie, Bobby$, 2, 5, 3, 8, 10\}$

  - $\{$Maggie, Bobby$\} \cup \{2, 5, 3, 8, 10\} \cup \{$red, green, blue$\}$
    $= \{$Maggie, Bobby$, 2, 5, 3, 8, 10,$ red, green, blue$\}$

  - $\{$Maggie, Bobby$\} \cup \{$Maggie, Bobby$\} = \{$Maggie, Bobby$\}$

- "Intersection" is $\cap$, and represents the set of elements in *both* sets

  - $\{$Maggie, Bobby$\} \cap \{$Bobby, Bogdan$\}$=$\{$Bobby$\}$
  - $\{$Maggie, Bobby$\} \cap \{2, 5, 3, 8, 10\} = \emptyset$

- $\emptyset$ is the "empty set" which is the set that contains nothing, or $\{\}$

- The output of union and intersection operations *is always a set*!
  - $\{$Maggie, Bobby$\} \cap \{$Bobby, Bogdan$\} \neq$ Bobby

- If two sets have an empty intersection, they are *disjoint*
  - $\{$Maggie, Bobby$\}$ and $\{2, 5, 3, 8, 10\}$ are disjoint

Oftentimes we don't want to (or can't) simply list all the items in a set, so we define sets in other ways (note we *always use the curly braces*)

- The set of all the boys' names that start with a vowel:
  $\{x : x \text{ is a boy's name starting with a vowel}\}$
- The set of all real numbers greater than -17.5 and less than 4:
  $\{x \in \mathbb{R} : -17.5 < x < 4\}$

Using this kind of notation, we can define the following:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

Claim: The empty set $\emptyset$ is a subset of every set.

Direct proof: First, note that $B \subset A$ is equivalent to the statement that

$$\forall x \in B \text{ we have } x \in A.$$

We want to prove that $\emptyset \subset A$ for any set A.

This is equivalent to saying that for all $x \in \emptyset$, we have $x \in A$. But we know that $\nexists x \in \emptyset$. Therefore the statement that these (nonexistent) x are in A is vacuously true. $\square$

Conditional statements & vacuous truths

Claims of the form $A \Rightarrow B$ are *conditional statements*

- "If A is true then B" is true
- A is the *antecedent* and B is the *consequent*
- $A \Rightarrow B$ is false *only when* A is true and B is false
- If antecedent A isn't true then the conditional statement is *vacuously true*

Other vacuously true statements:

Claim: If the moon was purple, then rain would fall up.

Definition: A binary relation is *quasitransitive* if $a > b$ and $b > c$ implies that $a > c$.

Claim: The binary relation with $a = b$ and $b > c$ and $a = c$ is quasitransitive.

Claim: The empty set $\emptyset$ is a subset of every set.

<u>Proof by contradiction</u>: First, note that $B \not\subset A$ is equivalent to the statement that

$$\exists x \in B \text{ such that } x \notin A.$$

Suppose that $\emptyset \not\subset A$. Then by the above statement we know that $\exists x \in \emptyset : x \notin A$. However, this contradicts the fact that $\nexists x \in \emptyset$. Thus, $\emptyset \subset A$. $\square$

More definitions

- A *family of sets* is a set whose elements are themselves sets
  - $\{\{1, 2, 4\}, \{2, 6\}, \{\text{green}\}\}$
- A well-known family of sets is the *power set* of any set X: it is the set of all subsets of X, and is written 2^X
  - $2^{\{\text{cat, dog}\}} = \{\emptyset, \{\text{cat}\}, \{\text{dog}\}, \{\text{cat, dog}\}\}$
- $|X|$ is the cardinality (size) of set X; the number of elements in X
  - If $|X| = n$ then $|2^X| = 2^n$

Quick aside on proof strategies:

Proof by contradiction: We want to show $A \Rightarrow B$. Assume that both A and $\neg B$ hold simultaneously. Show that this implies a contradiction.

Proof by contrapositive: We want to show $A \Rightarrow B$. Prove that $\neg B \Rightarrow \neg A$.

Proof by induction: We want to show that a property holds for all n, where n is an integer. (1) Prove it holds for $n = 0$ (or $n = 1$). (2) *Assume* it holds for arbitrary n. (3) Prove it must hold for $n + 1$.

Claim: If $|X| = n$ then $|2^X| = 2^n$

Proof by induction: Suppose that $|X| = 0$. If X has no elements, then $X = \emptyset$. It follows that $2^{\{\}} = \{\{\}\}$: the set of all subsets of $\emptyset$ is just the set containing $\emptyset$. Thus, $|2^{\{\}}| = 2^{|X|} = 2^0 = 1$, proving our statement for $n = 0$.

Suppose our statement holds for $|X| = n$, so $|2^X| = 2^n$.

Let $Y = X \cup \{x\}$, so that $|Y| = n + 1$. 2^Y is $2^X \cup \{S : S \subset Y \text{ and } x \in S\}$. Note that 2^X and $\{S : S \subset Y \text{ and } x \in S\}$ are disjoint.

The set $\{S : S \subset Y \text{ and } x \in S\}$ has 2^n elements: it is $\{S : S = \{x\} \cup Z \text{ where } Z \in 2^X\}$. Thus, $|2^Y| = 2^n + 2^n = 2 * 2^n = 2^{n+1}$. This proves our result. $\Box$

- If $A \subset B$ then the set of things that are in B but not A is the *complement of A in B*

$$B \setminus A = \text{ The complement of } A \text{ in } B = \{x \in B : x \notin A\}$$

- A set with a finite number of elements is a finite set
- A set with an infinite number of elements is an infinite set

Sets of numbers

- The real numbers, $\mathbb{R}$: All numbers that are positive, negative, or 0
- The natural numbers, $\mathbb{N}$: The set of all strictly positive integers
  $\mathbb{N} = \{1, 2, 3, 4, ...\}$
- The integers, $\mathbb{Z}$: the natural numbers, their negatives, and 0
  $\mathbb{Z} = \{..., -2, -1, 0, 1, 2...\}$

Ordered lists

- Unlike sets, *ordered lists* are denoted with smooth brackets (c, a, b, d, e)
  - Elements can repeat
- The 2-dimensional plane is represented by $\mathbb{R}^2$, or $\mathbb{R} \times \mathbb{R}$

$$\mathbb{R}^2 = \{(x, y) : x \in \mathbb{R} \text{ and } y \in \mathbb{R}\}$$

Cartesian product

- The general idea of an ordered pair doesn't require it to consist of two real numbers (or even two elements of the same set!)
- Given two sets X and Y, the Cartesian product of X and Y is $\{(x, y) : x \in X \text{ and } y \in Y\}$
  - $\{\text{cat, dog}\} \times \{a, b\} = \{(\text{cat}, a), (\text{cat}, b), (\text{dog}, a), (\text{dog}, b)\}$
- $X_1 \times X_2 \times ... \times X_n = \{(x_1, x_2, ..., x_n) : x_i \in X_i \text{ for } i = 1, 2, ..., n\}$
- $\mathbb{R}^n$ is the n-ary Cartesian product of $\mathbb{R}$

$$\mathbb{R}^n = \{(x_1, x_2, ..., x_n) : x_i \in \mathbb{R}\}$$

- An ordered list of numbers is a *vector*
- If the vector has n elements, then it is an element of n-dimensional Euclidean space, or $\mathbb{R}^n$
  - $(1, 5, 7) \in \mathbb{R}^3$
  - $(3, 2, 7, 12, 65, 78787, 3) \in \mathbb{R}^7$
- If $n \leq 3$, you can plot out the vector (as either on a line, the plane, or 3-D space)
- If $n > 3$ you can't plot the vector out, but there is no other conceptual difference

Functions

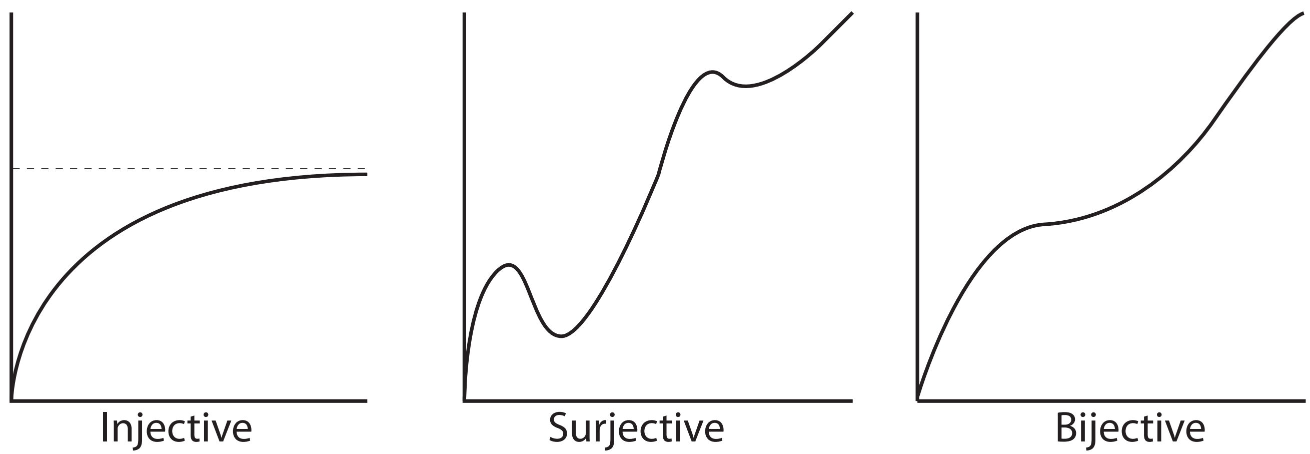
- A *function* is a mapping from a set X into a set Y
- This means that for each element of X, f assigns an element of Y, or $f : X \to Y$
- The *graph* of a function from $\mathbb{R}$ to $\mathbb{R}$ is the set

$$\{(x, y) \in \mathbb{R}^2 : y = f(x)\}$$

A function $f : X \to Y$ is *one-to-one* (or injective) if every element of X maps into a unique element of Y ($f(a) = f(b) \Rightarrow a = b$)

A function $f : X \to Y$ is *onto* (or surjective) if every element of Y is mapped into by some element of X ($\forall y \in Y, \exists x \in X : f(x) = y$)

A function that is one-to-one and onto is bijective



Composite functions

- Mappings can be combined so that if $f : B \to C$ and $g : A \to B$ then $f \circ g = f(g(x))$ is a mapping from A to C
  - $x \in A$, $g(x) \in B$, and $f(g(x)) \in C$

Example:

$f : \mathbb{R}^2 \to \mathbb{R}$, with $f(x, y) = x + 2y$

$g : \mathbb{R}^3 \to \mathbb{R}^2$, with $g(x, y, z) = (x^2 + 2y * z, 7)$

$$h = f(g) : \mathbb{R}^3 \to \mathbb{R}, \text{ with}$$
$$h(x, y, z) = x^2 + 2y * z + 14$$

In-class problems

3.1.2 Let A, B, C be sets such that $A \subset B$ and $A \cap C = \emptyset$. Simplify the following, if possible:

– $A \cap (B \cup C)$

– $(A \cap B) \cup C$

– $A \cup (B \cap C)$

– $(A \cup B) \cap C$

3.4.1 Let $f : \mathbb{R}^2 \to \mathbb{R}^2$. Interpret the mapping $f(x, y) = (x, -y)$ geometrically

- Which points remain unchanged under this mapping?
- Now answer both questions for $f(x, y) = (x, -4)$